

1 • Shared evidence: workshop cases

Change detectives: what became something new?

Name _____ Date _____

Constructed teaching cases. Use these for W1-W3; no practical experiment is required.

A • Ice tray

Ice warms into liquid water. Cooling below its freezing point makes ice again.

C • Iron bracket

After days in moist air, a brown coating forms. Analysis identifies rust, a new material.

E • Candle wax

A: solid wax melts. B: wax vapour burns with oxygen, forming carbon dioxide and water.

B • Paper ticket

A ticket is torn into pieces. Tests identify the pieces as the same paper material.

D • Wood offcut

Burning forms ash and gases. Cooling the products does not restore the wood.

Rule to audit

"If I cannot put it back easily, it must be a chemical change."

W1. Sort A-D: new substance / no new substance. Explain one choice.

W2. Repair the rule using case B.

W3. Classify E-A and E-B. Which change can cooling reverse?

2 • Supported route: diagnosis cards

Use page 1. Circle, label, draw or dictate. Word bank: oxygen / water / rust / new.

S1. Rusting makes ...

- A. A new material
- B. Only smaller pieces of iron

S2. Complete: Iron reacts with _____ and _____ to form _____ .

S3. Melting ice is a physical change because ...

- A. The substance is still water
- B. All changes involving heat are chemical

S4. Explain why tearing paper does not prove a chemical change.

S5. Write two candle labels: one for melting and one for burning. Include “new substance” in one label.

3 • Standard route: repair report

Use evidence, not just appearance. A short sentence or labelled diagram is enough.

Case	Physical or chemical?	Evidence
Ice melts		
Paper tears		
Iron rusts		
Wood burns		

N1. Complete the classification table.

N2. Explain why rusting is not reversed by simply cooling the bracket.

N3. A visitor says wax melting and burning are the same. Write a two-sentence correction.

N4. Explain why some products of burning are missing from an open tray.

4 • Stretch route: test a tempting rule

Two claims need careful correction. No chemical equation is required.

Claim A: “Every change that is hard to undo forms a new substance.” Claim B: “Only 2 g of ash remained from 10 g of dry wood, so 8 g of matter was destroyed.” These masses are a constructed teaching example from an open system.

T1. Use a counterexample to evaluate claim A. Give a better classification rule.

T2. Explain claim B’s error. Include gases and oxygen from the air. Why can we not calculate the total gas mass as simply 8 g?

T3. Design two contrasting exhibit labels for wax melting and burning, explaining the role of cooling.

5 • Exit and reflection

Work independently. Use science words where helpful.

E1. Explain why rusting is a chemical change.

E2. Why can tearing paper be a physical change even if it is difficult to restore?

E3. Give one difference between wax melting and wax burning.

E4. Where can some burning products go in an open system?

Reflection. Which workshop claim would you now challenge, and why?
